

An Investigation into the Influence of Embodied Agents on the Social Presence Between Humans in Virtual Reality Rowing

Frederique Voskeuil

Interaction Technology
University of Twente
Enschede, Netherlands
frevoskeuil@gmail.com

Dees Postma

Human-Media Interaction
University of Twente
Enschede, Netherlands
d.b.w.postma@utwente.nl

Armağan Karahanoğlu

Interaction Design Group
University of Twente
Enschede, Netherlands
a.karahanoglu@utwente.nl

Dennis Reidsma

Human-Media Interaction
University of Twente
Enschede, Netherlands
d.reidsma@utwente.nl

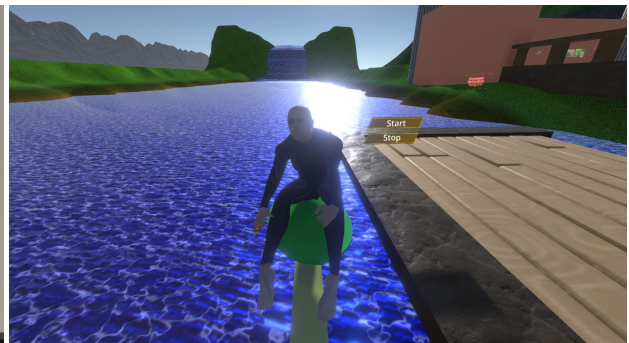


Figure 1: Left: 2 people on ergometer. Right: a coxswain addressing the rower in the back of the Virtual boat.

Abstract

This paper investigates the relations between the embodiment of virtual agents and the sense of social presence in virtual reality (VR) for rowing simulation. We carried out observations and an experimental study to investigate the role of embodied agents in social presence. We developed an embodied coxswain and set up an experimental study with two RP3 rowing machines, VR headsets, and various hardware and software components. Twenty-two participants tested the system under two conditions: with an embodied coxswain and with a non-embodied agent. Experiment results showed a significant increase in perceived social presence in the VR between participants when rowing together with the embodied coxswain. Our findings illustrate the challenges and benefits of more realistic coxswain representations in VR rowing and pave the path for further exploration into the virtual agent characteristics in other sports that can enhance social presence for colocated participants in VR settings.

CCS Concepts

- **Human-centered computing** → **Empirical studies in collaborative and social computing**; *User studies*; *Virtual reality*;
- **Software and its engineering** → **Virtual worlds training simulations**.

Keywords

SportsHCI, virtual rowing, virtual reality, social presence

ACM Reference Format:

Frederique Voskeuil, Armağan Karahanoğlu, Dees Postma, and Dennis Reidsma. 2025. An Investigation into the Influence of Embodied Agents on the Social Presence Between Humans in Virtual Reality Rowing. In *Annual Conference on Human-Computer Interaction and Sports (SportsHCI 2025)*, November 17–19, 2025, Enschede, Netherlands. ACM, New York, NY, USA, 11 pages. <https://doi.org/10.1145/3749385.3749390>

1 Introduction

In recent years, virtual collaborative sports have attracted growing attention within HCI, and particularly in SportsHCI. These activities allow people to exercise in digitally mediated environments, from running with fitness apps such as RunKeeper¹ or Strava², to more immersive virtual experiences that use Augmented or Virtual Reality (VR) headsets or haptics. Many of these systems integrate



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SportsHCI 2025, Enschede, Netherlands

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ACM ISBN 979-8-4007-1428-3/25/11

<https://doi.org/10.1145/3749385.3749390>

¹Runkeeper – <https://runkeeper.com/cms/>

²Strava – <https://www.strava.com/?hl=nl-NL>

social features to keep users engaged. For instance, Zwift³ enables indoor cyclists to train and compete in an online community, while iFit⁴ offers interactive running, cycling, and yoga classes. In such settings, a strong sense of social presence, feeling connected and “with” others, is often central to a positive experience [21, 31, 60].

Social presence holds the potential to enrich many facets of virtual collaborative sports experiences, such as improving performance and attracting broader user participation [45]. While the influence of social presence in virtual collaborative sports experiences has been researched in the fields of cycling [3, 11], volleyball [71], jogging [47, 50], boxing [42] and other sports [43], it remains a compelling topic to explore further, due to many unexplored impacts on performance and joy of sports. For example, virtual collaborative experiences offer both opportunities and challenges for team sports like rowing, which rely on social inter-personal coordination and social interaction among team members.

With this context in mind, the primary aim of this paper is to explore how to enhance social presence between team rowers in a Virtual Reality (VR) rowing session by employing a virtual coxswain. The study is motivated by the aspiration to unlock the potential of embodied coaching to enhance team dynamics within rowing teams and aims to contribute to the realm of virtual collaborative sports experiences. Such systems have been built and investigated in the past [e.g. 16, 48, 73]; we build on a similar setup, while the team setup and the ways to foster social presence have not been addressed yet.

In this paper, we aim to explore how to enhance a social presence between rowers in a VR setting. We do this by introducing a virtual character acting as a coxswain (i.e. the person steering the boat, managing pace and race strategy, and coaching the other crew members) that interacts with both the rowers together, explicitly addressing them in a multi-party multi-modal interaction manner. After grounding this design in preliminary observations, we show through an empirical study that the presence of this virtual character with its behavioural strategies increases perceived social presence *between* the rowers themselves in a team rowing task. Given the interpersonal dependencies that play a role in team rowing, we argue that such increased perceived presence may have a positive impact on learning and performance in VR-based training setups like our system.

Thus, our research question is: *How can social presence between two rowers during a Virtual Reality rowing session be enhanced by employing embodied virtual agents?* By addressing this question, we contribute to the state of the art in the design of Virtual Reality sports experiences. In the following lines, we will first explain the context of rowing, the current state of SportsHCI for rowing and social presence in VR. Following, we will explain our participant studies in which we addressed our research question.

2 Related Work

Rowing is an on-water activity, which requires athletes to use their muscle power and technique in “rowing strokes”. A rowing stroke consists of four phases: the catch, the drive, the finish, and the recovery. The relative timing of these phases *within* and *between* crew

members of a boat is key to performance in on-water rowing. Often, crew rowers are supported by a coxswain. The coxswain is seated in the bow-facing direction and is the only boat member who does not row. They are responsible for steering the boat and communicating with the rowers. Besides, a good coxswain motivates the rowers and ensures the game plan of the coach is executed.

2.1 Rowing on Land

While rowing is an outdoor sport and requires specific equipment for training (e.g., the boat) and training conditions (e.g., still water), weather conditions can sometimes restrict training on water, making indoor rowing on a rowing machine an integral part of this sport. Ergometers and virtual reality systems stood out in the last decade as alternatives to in-water rowing training to tackle the challenges stemming from seasons and weather conditions.

Several commercial systems enhance the on-land virtual rowing experience. For example, Holofit⁵ is a VR rowing fitness app that users can download onto their VR headset. In rowing exercises, the user can explore different environments from real-world or imaginative locations while rowing. Another example is Quiske⁶ which employs motion-tracking devices to provide real-time feedback to rowers about their performance. Similarly, BioRowTech⁷ provides equipment and technology for optimizing rowing performance, which provides visual feedback to rowers about common errors, like incorrect arm movements, bending the back too early, and excessive back movement at the finish. Finally, the M3 (Multi-Modal Motion synthesis) simulator⁸ offers an immersive rowing experience which combines haptic oar movement with visual and auditory feedback, and guides the path of the oar of the rower. All these examples help rowers improve their technique, efficiency, and overall performance.

In research systems, technology has been used in various ways, mostly for exploring the role of feedback in learning and performance, with VR as well as without it. [2, 19, 48, 63, 65–67, 73]. Adaptive, context-aware feedback has been shown to increase engagement and training effectiveness in VR settings [18, 23]. By linking agent responses to real-time biometric or performance data, designers can create a more credible and personalised coaching experience, which our findings suggest would further enhance social presence. While existing systems and prior studies investigated the technological aspects of virtual rowing and the role and effect of feedback in various modalities, there is no study yet which investigates the social presence for multiple participants in VR rowing as an integral part of rowing. In the next section, we will tackle the definitions of social presence which forms the core of our investigation.

2.2 Defining Social Presence in VR

Social presence is a multifaceted construct which is the degree of awareness or connectedness with other users in a (virtual) environment [25]. It contributes to the overall experience of exercising in virtual environments, as it enhances enjoyment and engagement

³Zwift – <https://us.zwift.com/>

⁴iFit – <https://www.ifit.com/connected-fitness>

⁵Holofit – <https://www.holodia.com/>

⁶Quiske – <https://www.rowingperformance.com/>

⁷BioRowTech – https://biorow.com/index.php?route=product/product&path=61_108&product_id=55

⁸Multi-Modal Motion synthesis) – <https://www.rowing.ethz.ch/>

[34, 70], fosters friendships [44] and can even increase the efficacy of training and therapy programs [70]. Moreover, the perception of presence generally improves learning, efficiency, planning, and cognitive or sensorimotor performance, and facilitates the transfer of training to real-world situations [29].

Prior work on avatar-based social interaction highlights the role of embodiment in shaping presence and rapport [5, 28, 49, 59]. These studies help us understand how anthropomorphism, agency cues, and sensory fidelity interact to influence perceived co-presence, providing a foundation for interpreting our findings. Meanwhile, several aspects affect the perception of social presence in real and virtual environments (i.e., space and interactions, perception and understanding and emotional connectedness). These aspects will provide us with a grounding for addressing the research question that we will tackle through a set of studies.

2.2.1 Space and Interactions. Social presence is often characterized as the experience of being in one place, while physically being located in another [27, 37, 52, 69, 76]. It is a feeling of being present or in another world without mentioning the transition from the physical world to a virtual world [25, 34, 70, 71] or a computer-simulated environment [29]. Yet, social presence is feeling present in a virtual environment while physically being located in another one [25, 37, 70, 71, 76]. The sense of social presence is further shaped by interactions, [10, 11, 25, 26, 37, 42–44, 47, 51, 70, 74], especially with other (virtual) humans [10, 11, 25, 42–44, 46, 47, 50, 70, 74]. All these prior works highlight the importance of the social context, including terms like social play, sense of community and social interaction, for the perception of social presence.

2.2.2 Perception and Understanding. Harms and Biocca [26] express the need to understand other (virtual) humans in addition to the exchange of social cues. They propose that presence can be defined by the degree to which the user understands messages, and the emotional and attitudinal state to and from other users. Yet, understanding messages to and from others in virtual environments is an important element of social presence [28]. These highlight that the awareness of the presence of other (virtual) humans and understanding messages to and from these humans defines the user perception and understanding of presence.

2.2.3 Emotional Connectedness. Emotional connectedness is subjective and characterized by the relationship between the user and other (virtual) humans. Consider for instance that people show greater (emotional) connection and affection to avatars that share great resemblance to the user as opposed to avatars that share little resemblance [6, 7]. It includes the degree to which users feel united with others. Consequently, social presence is the perception of being together with others [25, 34, 70]. Moreover, some prior work draws a connection between emotional states and social presence [26], which express that social presence not only influences the emotional and attitudinal state of others but also affects the behaviour of each other. For example, social presence relies on the degree of warmth, sociability, personalization, sensitivity, or intimacy of a medium when it is employed to interact with others Lombard et al. [37], Nunez et al. [51]. Conclusively, (emotional) connectedness entails the subjective experience of being with or feeling (emotionally) connected to other (virtual) humans.

In summary, the prior work shows that social presence is highly subjective and necessary to acquire to enhance the virtual experiences, while the absence of the feeling can distract the experience. Considering the importance of togetherness in team sports, we will address how to enhance the “togetherness” in VR rowing.

3 Design for VR Social Presence in Rowing

We carried out (1) a preliminary observation and (2) an experimental study to address our research question. The preliminary study mainly served to inform our design of the embodied virtual coxswain as a means to influence the perceived social presence between two team rowers in VR; the experimental study explored its actual effects. We obtained ethical approval from our institute before contacting any participants and obtained their informed consent.

3.1 Preliminary Observations

To gain deeper insights into the behavioural and social dynamics in crew rowing, we carried out semi-structured observations at a local rowing club. The first author investigated the behavioural and social dynamics in a (coxed) eight training sessions, consisting of eight male rowers, a coxswain, and a coach (aged 18–25). The training sessions took place along the local canal. Additionally, we performed an unstructured interview with the coach after the training. In addition to on-site observations, we made semi-structured observations on video-recorded rowing matches to improve our understanding of team dynamics in rowing. We selected videos featuring Olympic rowing teams, as these matches are often well-recorded and have a high video quality [53–58, 64]. The rowers in the videos were male and female Olympic rowers. The dataset included videos of two men’s eight boats, one men’s two boat, three women’s eight boats, two women’s two boats, and a video specifically of a coxswain during a match. Olympic rowing teams, composed of professionals, are expected to present strong team dynamics. In our notes, we documented the (relative) occurrence of various social and behavioural events, such as ‘high fives’ and other (social) interactions. The notes of the author were then discussed with the research team to decide upon the important elements of social presence in VR rowing.

We noticed that only the coxswain and the coach used verbal communication during the training. The coxswain spoke in a rhythmic pattern and took the role of the coach within the boat. Additionally, the coach used a megaphone to communicate instructions to the coxswain. Notably, the rowers interacted very little with each other during rowing because they were often too exhausted to talk. They communicated a lot with each other pre- and post-training – these interactions were often casual in nature. A strong sense of team spirit was evident within the team. We observed that the team coach actively provided feedback during intermediate discussions. However, the rowers did not communicate much during rowing matches, while they did engage in casual interactions in pre-matches.

In Olympic rowing match videos, we uncovered several shared experiences such as shared celebrations and shared grief. The videos showed that rowers often use non-verbal communication when they win, such as high-fives, handshakes, and shoulder taps. The

celebration forms of the male rowers showed a wide range of diversity [55, 58, 64]. Especially shoulder taps (observed four times) and hugs (observed three times) were the most frequently observed methods, followed by handshakes (observed two times), sometimes even with opponents (observed two times), and taps on other body parts, such as the head or knee (observed two times). In contrast, in the videos, female rowers primarily hugged when celebrating [53, 54, 56, 57]. Male rowers tended to tap on each other's knees during moments of grief (observed two times), whereas there was almost no interaction between female rowers [55]. Only an occasional tap on the knee is observed (observed once) [53]. When the rowers experienced victories or losses in the race (as part of training), there were limited occurrences of shared celebrations or grief among the rowers. This entails a variety of verbal or non-verbal interactions between the rowers as a result of them winning or losing.

An analysis of the coxswain's video [62] makes it apparent that the coxswain communicates a lot, especially in contrast to the rowers who minimally engage in communication. It is observed that a coxswain uttered rhythmic and motivational sentences, which were different every time. The coxswain constantly reminded the rowers how important it is to win and how hard they have trained. Notably, the coxswain occasionally gave a rhythm, especially during pace adjustments, and to remind the rowers to manage their rhythm. Moreover, the coxswain addresses the rowers as a team and keeps them informed on the positions of other boats. Lastly, the coxswain gave updates on the distance until the finish line.

3.2 Development of VR Coxswain for Enhancing Social Presence

Our preliminary observations demonstrated that (emotional) connectedness and social interaction are the two important aspects of social presence. Our findings also demonstrated that the rowers do not communicate with each other but with the coxswain during training and race. By building on these findings, we decided to create a *scripted* virtual coxswain and investigate the effect of virtual coxswain in social presence in VR rowing.

Based upon our deliberate decision, we developed a virtual coxswain, by using motion tracking technology (Tundra Trackers) in combination with the Metagen AI platform⁹. We used the physical input to animate our virtual coxswain. To achieve motion capture, we used trackers which were positioned at the chest, hips, and feet of a male individual. Additionally, we used a headset and controllers to create a life-like animation of the virtual coxswain (see Figure 2). In addition to a realistic appearance, the coxswain should have a realistic voice, therefore, A 21-year-old male provided the voice to our coxswain.

To achieve a high level of realism to the coxswain, we used face tracking through the Vive Facial Tracker. The Vive Facial Tracker¹⁰ can record up to 38 different facial movements and can be easily attached to the Vive Pro Headset (see Figure 3). Face tracking allowed us to develop the virtual coxswain to make eye contact with the rowers and to show appropriate facial responses when communicating with the rowers.



Figure 2: The movements captured in real-time and translated into a virtual avatar



Figure 3: The figure on the left shows the coxswain reproducing the movement of an actor. The figure on the right shows the real-time facial tracking from the Vive Facial Tracker

Our system tracks the animation of the face and gestures of the virtual coxswain as well as the positions and movements of the rowers on the rowing machine in real time and maps these to animations in the used VR platform (NeosVR). Furthermore, to keep the experiment flowing smoothly, we implemented ways to start, stop, and pause animations to deal with unexpected events during a session. First, the animations are tested on a free character from the NeosVR library. This character is already equipped for face tracking. After considering the number and positioning of the trackers, animations with this character are recorded and reviewed. Once the animations seem satisfactory, a more suitable avatar is searched. Multiple characters are inspected, and a male, sporty character is eventually chosen. This character contains a mesh, textures, and a rig (including the face) and is ready for animation.

4 Testing the Effect of Virtual Coxswain on Social Presence

Our experiment aimed to test whether a virtual coxswain would enhance the social presence between two rowers during a Virtual Reality rowing session. In our experiment, we asked participants to engage in a virtual rowing task consisting of two conditions: participants were guided by a non-embodied coxswain, and participants were guided by the embodied coxswain we described in section 3. Our goal was not to validate that a virtual coxswain would be the ultimate solution for the challenge we are tackling, but more to explore the ways to address it. Hence, we see our experiment as a research through design experiment [22, 77] which aimed to produce knowledge rather than testing a solution. Our experiment was approved by the ethics board of our institute, and informed consent was obtained from the participants.

4.1 Participants

The participant pool consists of 22 individuals, of which 14 are males and 8 are females. The ages of the participants ranged from 19 to 25 years old, and they were all students at our institute living in the same town. Additionally, 21 participants had the local nationality

⁹Metagen – <https://github.com/MetaGenAI/MetaGenNeos>

¹⁰VIVE Facial Tracker – <https://www.vive.com/accessory/facial-tracker/>

and spoke the local language as their native language whereas one participant exclusively spoke English. Prior knowledge of rowing or Virtual Reality was not a requisition for the participants. Lastly, all participants were physically able to join the rowing exercises.

4.2 Measurements

We obtained different measurements to investigate social presence in VR rowing. We used the Networked Minds Measure Questionnaire by Harms and Biocca [26] to assess the sense of social presence between the members of a rowing dyad. The Networked Minds questionnaire concerns social presence and consists of six sub-dimensions (co-presence, attentional allocation, perceived message understanding, perceived affective understanding, perceived affective interdependence, and perceived behavioural interdependence). The reliability of the original questionnaire is high with Chronbach's Alpha ranging between .81 and .87 for the sub-scales. To assess the perceived level of anthropomorphism of the non-embodied agent and the embodied coxswain, we used a combination of the ASA questionnaire [20] and the Rapport Scale [1]. The ASA scale is an instrument for evaluating human interaction with an artificial social agent and the Rapport Scale offers a measure of cognitive rapport. This allowed us to discuss whether our design of the coxswain behaviours was appropriate to their task.

Finally, we engaged in a post-experiment discussion with participant dyads. For our discussion, we prepared a number of open-ended questions (e.g., What did you like about the rowing exercise in condition n"; what did you dislike about the rowing exercise in condition? and why?). Further, we asked participants to reflect on the nature of the presented agents through a number of pictures (adapted from Charisi et al. [15], see Figure 4) prompting them to explain which picture would best fit the nature of the virtual agent.

4.3 Experiment Design and Procedure

To examine the effect of avatar realism on the sense of social presence, we carried out a within-subjects experiment in which the participants were invited to test two conditions: 'non-embodied' and 'embodied'. In the non-embodied condition, participants were asked to engage in a virtual reality rowing session which was guided by a non-embodied agent (Figure 5). Guidance from this agent came in the form of verbal instructions that were also visible on a virtual pane. In the embodied condition, participants were instructed by a virtual embodied coxswain and asked to perform the same rowing tasks (Figure 5). All participants experienced both the non-embodied and the embodied coxswain conditions. To eliminate order effect in the results, we randomised the order of the conditions between rowing dyads. For the experiment, we used the networked virtual reality rowing setup. Each of the ergometers was placed in a different room and participants were connected through the audio-visual feedback that was relayed to them through the virtual world.

Before the start of the experiment, participants were randomly paired and randomly assigned a seating position in the boat (i.e. front-seat or back-seat). They were then asked to indicate their level of perceived closeness using the IOS-Scale [4]. After completing the first condition, participants were asked to fill out the Networked

Minds Measure Questionnaire [26], the Rapport Scale [1], and the ASA Questionnaire [20] – see also 'Measurements section' below. After completion of the second condition, participants were again asked to fill out the same questionnaires. The experiment ended with post-experiment interviews.

In our setup, participants alternated between bow and stroke positions. The bow seat faces the coxswain directly and is typically responsible for synchronising with the stroke seat, who sets the rowing pace. This positional difference may have affected the salience of the coxswain's gestures and gaze direction.

4.4 Analysis

Data from the questionnaires were analysed using mixed effects regression [see e.g., 13, 17, 75]. With mixed effects regression, we are able to account for nested dependencies in the data. In the present case, such nested dependencies arise because two participants form a dyad, meaning that the data points from one participant are not truly independent. Data from the Networked Minds Measure Questionnaire and the combined ASA/Rapport questionnaire were analysed separately. Shapiro-Wilk's test for normality indicated that the assumption for normality was met for both the Networked Minds questionnaire ($p=0.27$) and the combined ASA/Rapport questionnaire ($p=0.77$). Data were analysed using the *lme4*-package [8] and the *lmerTest*-package [32] in R-statistics [61]. For both questionnaires, we constructed a regression model with the summed questionnaire data as the dependent variable, condition and seating position as the main effects, and an interaction effect for condition and seating position. To model the random-effects structure, we allowed model intercepts to vary per dyad. Finally, a Bonferroni correction was applied to account for alpha inflation due to multiple testing.

The post-experiment discussions were analysed using an inductive thematic analysis approach [14]. Notes from the observations and post-test discussion findings were categorized and grouped into similar topics using the thematic analysis method proposed by them. Following this, reoccurring patterns or trends were identified, interpreted, and analysed.

4.4.1 Limitations. Some technical issues emerged throughout the experiments. During three experiments, audio-related issues arose, causing the participants to be unable to hear each other through the VR headsets. This occurred during experiments one, two, and five. Although the participants could hear each other via mobile phones, one participant noted that she could not hear the other person due to the noise of the rowing machine. During the other experiments, the participants could hear each other via the VR headsets. Another technical issue arose during the first experiment when rowers were unexpectedly thrown out of the (virtual) boat during their rowing session. They were placed back in the boat after which the experiment continued.

5 Results

5.1 Perceived Presence in Virtual Rowing

Mixed effects regression analysis of the Networked Minds questionnaire for social presence revealed that there was a significant

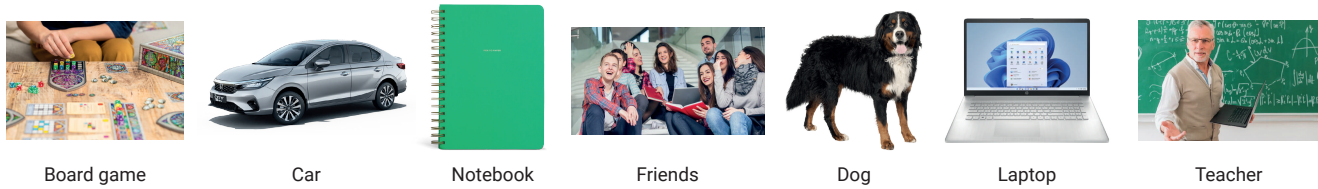


Figure 4: Pictures used in the post-experiment discussions, adapted from Charisi et al. [15] to fit an adult audience

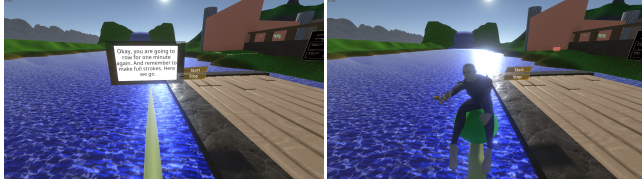


Figure 5: The two conditions of the experiment: a human-like coxswain (right) and a text-based form of feedback (left).

effect of condition ($p = 0.02$) on the sense of social presence between members of a rowing dyad (Figure 6). Participants scored higher on the Networked Minds questionnaire ($M = 0.23$; $SE = 0.07$) when the avatar was presented as an embodied coxswain than as a non-embodied entity. In the regression model, we also tested the effect of seating position on social presence. It might have been the case that a sense of social presence (towards the other rower) was related to seating position (e.g., being able to see the other participant, or not). This was, however, not the case, as we found no significant effect of seating position ($p = 0.72$) or interaction effect of seating position and condition ($p = 0.68$).

Data from the combined ASA / Rapport Scale for anthropomorphism were also analysed using mixed effects regression. It was found that there was a significant effect of ‘condition’ ($p < 0.001$) on the perceived level of anthropomorphism (Figure 6). Participants perceived the level of anthropomorphism to be significantly higher ($M = 0.84$; $SE = 0.18$) for the embodied agent than for the non-embodied agent. In the model, we also tested the effect of seating position on the perceived level of anthropomorphism. It might have been the case that the virtual seating position in the boat obscured a clear view of the agent, lowering the differences between the two types of agents. This was, however, not the case, we found no significant effect of seating position ($p = 0.67$) or interaction effect of seating position and condition ($p = 0.82$).

5.2 Results of post-experiment interviews

During the post-experiment discussions, we found that $N=13$ participants referred to the non-embodied agent as “it”, while the other participants referred to it as “he”. Conversely, it was the other way around for the embodied agent, as the majority of the participants often referred to the embodied agent as “he” ($N=17$), while five participants addressed the avatar with “it”. We will elaborate our findings in the following sections.

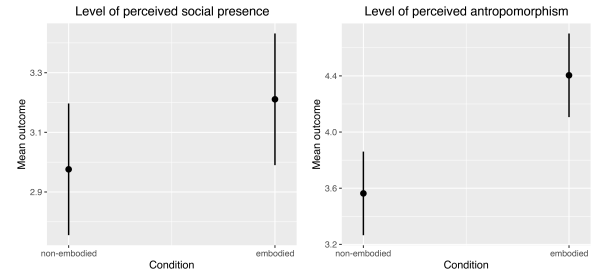


Figure 6: Partial effect plots for ‘condition’ for the level of perceived social presence (Panel A) and for the level of perceived anthropomorphism (Panel B). Levels of condition (i.e. non-embodied / embodied) are on the x-axis; the grand-mean centred values (± 2 SD) for the respective questionnaires are on the y-axis.

5.2.1 Perceptions about Non-Embodied agent. When looking at how the participants perceive the agents, the non-embodied agent was often described as programmed ($N=11$), robot-like ($N=11$), and emotionless ($N=8$), accordingly linking the picture of the laptop to the agent ($N=20$). One of the participants stated: “It was just like an instruction system, and you see that often with laptops. It also didn’t have a face, it wasn’t a person” (experiment 8, back-seat rower). Nevertheless, $N=8$ participants noted that the agent gives clear instructions.

Meanwhile, almost half of the participants thought the agent gave clear instructions ($N=8$). One participant noted: “I think this agent was much clearer than the other agent, but it was boring” (experiment 2, back-seat rower). Still, some participants noted that the agent’s feedback was not directly related to the actions of the user ($N=3$). One participant said: “He said, ‘you are doing all right’, but I know it is not because he is triggered by something that happened. It’s not because he really understands what you do, I think” (experiment 9, back-seat rower). Furthermore, $N=4$ participants thought the agent had similarities with the notebook, while five participants thought it resembled a car, particularly a navigation system. One participant stated: “It looked like a navigation system that also tells you to go left here, go right now” (experiment 3, back-rower). Finally, two participants associated the agent with images of a teacher or a board game.

Nearly all participants pointed out that the agent had the least similarities to a dog (77%), as the participants stated that the system lacked emotions (36%), a characteristic that is often associated with dogs. One participant stated: “It doesn’t look like anything that lives.

A dog is always cute, fun or happy, and an AI doesn't have any emotions, it is emotionless" (experiment 10, front-rower). Another participant agreed: *"Dogs are very nice, spontaneous and sweet and the laptop wasn't"* (experiment 7, back-seat rower). In addition to a dog, almost half of the participants pointed out that the agent did not resemble a friend. One participant expressed: *"I think a computer screen to be the farthest from a friend"* (experiment 5, front-seat rower). Further, another participant described the interaction with the agent as being static: *"I would say the least similarities with friends. It was a somewhat static interaction, which is not the case with friends"* (experiment 4, front-seat rower). Lastly, a minority of the participants mentioned that the agent did not have any resemblance to a laptop (13%), a board game (9%), or a notebook (5%).

5.2.2 Perceptions about the Embodied Agent. The embodied agent is often characterised by his friendly voice (N=10), kind tone (N=10), and good feedback (N=5), which yielded participants' perceptions of the embodied agent as similar to a teacher (N=16) or a friend (N=9).

On the other hand, nine participants noted that the embodied agent had a strange placement and movements, and N=8 mentioned that the feedback of the agent was not a result of the rower's actions. N=5 participants noticed that the agent was boring due to the voice. One participant described the embodied agent as "monotone" and "robot-like" (experiment 4, back-seat rower). The voice of the agent played a role in this perception, as one participant pointed out: *"It was a very computer-like voice, so you associate that with robots, laptops, computers, and such. A mechanical voice"* (experiment 2, front-seat rower).

Participants associated the agent with being a teacher (N=16), while two of them thought the agent lacked the authoritative traits expected from a teacher. N=9 associated the agent with the "friends" and described the agent as being nice or friendly (N=6). One participant stated: *"His voice was very friendly and he gave a lot of compliments"* (experiment 10, back-seat rower).

Yet, some participants considered the agent to be flawed, which added a human-like feature. One participant noted: *"Sometimes he had to correct himself, and therefore I would say friends instead of a teacher"* (experiment 1, back-seat rower). Additionally, multiple participants sensed there was something off about the embodied agent (N=9). A participant expressed: *"He tried to level with us. I would think he resembled a friend, not your best friend, but he tried to be"* (experiment 6, front-seat rower). The movements and placement of the agent were perceived as strange (N=9), and three participants were even frightened of the agent. One of the participants stated: *"I thought he was human-like, however there was something off about the face and his posture. His voice was very human-like and nice to listen to, but the rest of the appearance wasn't real. Therefore, I thought he was scary"* (experiment 8, back-seat rower). Two participants liked that the agent was pointing at them. Additionally, one participant would prefer the agent to call him by his name.

Due to the perceived odd facial expressions and posture, four participants stated that the agent resembled the picture of the laptop. Additionally, 36% of the participants thought that the feedback of the agent was not related to their actions. For example, one participant stated: *"It would be good if he could give specific points you can recognize in your actions. Maybe that would make him more*

human-like, or human-like by being interactive" (experiment 3, front-seat rower). On the other hand, four participants did not realize that the embodied agent was pre-programmed. Three of these four participants thought that the responses of the agent were very well timed, and therefore it seemed like the agent responded to them. A participant asked: *"The coxswain was giving specific feedback for only one person at the time, right?"*. When she heard that the embodied agent was scripted, she noted: *"He gave feedback for one person at the time, so that's why we thought that he sees differences between the first and the second person"* (experiment 11, front-rower). Finally, six participants commented the embodied agent gave clear instructions.

According to the participants, the agent had the least resemblance to a notebook (N=8). One participant articulated: *"The notebook has the fewest similarities because that is the least human-like and interactive thing here"* (experiment 9, front-seat rower). In a separate experiment, a participant remarked: *"I had more of a teacher on board than a textbook"* (experiment 3, front-seat rower). Besides, four participants noted the agent is least similar to the car. A participant claimed: *"The car, notebook and the other things are not interactive and the coxswain is interactive"* (experiment 2, back-seat rower). Another participant agreed by expressing that a car operates the other way around, as the driver controls it, while in this scenario, the agent guides the participant (experiment 2, front-seat rower). Lastly, participants indicated the lowest level of resemblance between the agent and a board game (N=3), a dog (N=3), a teacher (N=2), friends (N=1) and a laptop (N=1).

5.2.3 Personal Preferences. More than two-thirds of the participants preferred the condition with the embodied agent (68%). Four participants compared the two agents and observed that the embodied agent had higher levels of motivation, enthusiasm, and engagement. In contrast, three participants favoured the condition with the non-embodied agent, as they noticed that the embodied agent could be distracting. One participant expressed this as: *"I didn't know where to look, at him or to still focus on the side of the boat to see the oars behind me"* (experiment 11, front-seat rower). Yet, it was perceived to be fun by the front-seat rower of experiment 8: *"As soon as the coxswain started talking for a longer period of time, I found it a bit distracting. I responded to it, which is fun, but it does distract a bit from the task, rowing"*. Another participant indicated being more focused on the environment when rowing with the non-embodied agent (experiment 9, front-seat rower). On the other hand, back-seat rower of experiment 5 disagreed with this, as he was too busy reading the instructions during the non-embodied agent condition.

6 Discussion

Prior research highlighted the effectiveness of virtual characters on motivation and performance levels [24]. Studies by Li et al. [36] and Hoffman et al. [29] demonstrate that rowing with immersive VR improves performance, motivation levels, and energy management. Moreover, Mouatt et al. [41] and IJsselstein et al. [30] suggest that immersive VR can enhance enjoyment and engagement in exercise compared to non-immersive VR and without VR. Building upon these studies, our goal in this paper was to discover ways to enhance the sense of social presence between human rowers. Our results

showed that participants generally liked the embodied agent, and it seemed that the embodied agent was perceived as more social than the non-embodied agent. From our findings, three points of discussion emerged about the experience of virtual coxswains in VR rowing: (1) the use of virtual humans to modify the perceived interaction and relation between human users in virtual reality; (2) the experience of the social agent characteristics; and (3) the value of using virtual humans for other purposes in VR rowing and other sports. Finally, we point out drawbacks that we observed regarding the behaviours that we designed for our virtual coxswain.

6.1 Virtual Humans Change the Interaction Between Real Humans

Our findings signal that introducing an embodied coxswain significantly enhances social presence *between* human team rowers in a VR rowing session. While rowing with an embodied coxswain appears to enhance social presence, it is also uncertain which characteristics of the coxswain cause this change. Our study did not have evidence which characteristics of the embodied coxswain have the most impact on social presence. It is conceivable that other characteristics of the athlete's interpersonal relation may similarly be influenced through the well-designed use of virtual characters.

To further unpack what elements of embodiment contribute (most) to the sense of social presence, future research might specifically investigate key aspects of nonverbal communication (e.g. gesturing, gaze behaviour, and facial expression [see e.g., 6]); avatar appearance [7]; and proxemics [21, 31, 60]. In doing so, it is crucial to also control for possible confounders such as the avatar's responsiveness, adaptivity, and emotional intelligence. While it is impossible to test – and control for – all of these factors simultaneously, factorial designs be helpful in further charting the realm of virtual avatars and social presence in sports.

Yet, our findings were reinforced through the findings of the LMER analysis of the Networked Minds Measure questionnaire. A virtual coxswain may, thus contribute to those facets of VR rowing training where presence plays a role. For example, in team rowing, interpersonal coordination is important and rowers in a team should be very attuned to each other through behavioral and social cues. Other sports may have similar characteristics, and may similarly benefit from using virtual characters to influence the relation between human athletes that are together in VR.

Although our study centred on the coxswain–rower relationship, rowing is inherently collaborative between rowers. Future work could explore how embodied agents might facilitate inter-rower synchronisation, possibly by mediating feedback or enhancing non-verbal coordination cues (e.g., [10]). Besides, given that most participants reported little or no prior VR experience, it is plausible that part of the observed social presence effect reflects a novelty response to immersive technology [39]. Therefore, a longitudinal evaluation would help determine whether the impact of embodiment persists after repeated exposure.

6.2 Effects of Social Characteristics of the Virtual Human

In our experiment, we introduced an embodied coxswain as we thought that through its social characteristics, an embodied coxswain

could contribute to the relationship between human rowers. Indeed, the embodied agent seemed to have more social characteristics than the non-embodied agent, which was the expected result of our experiment. We found that our participants frequently expressed having a different emotional connection to the embodied agent compared to the non-embodied agent, as the embodied agent was often perceived as more friendly, positive, and engaging. It also seemed that the embodied agent was experienced as more of a *person* (seen LMER analysis of the Rapport Scale and ASA questionnaire and the analysis of the post-experiment discussions). These mean that our manipulation achieved what we expected. Still, some participants reported that the coxswain's feedback was not contingent on their rowing performance point to a conceptual, not merely technical, limitation. In social presence theory [12] contingent, context-relevant behaviour is central to sustaining credibility and mutual engagement. Without such responsiveness, perceived authenticity of the interaction is diminished.

Our results also showed that rowers describe the embodied agent as friendly and enthusiastic, linking it to a teacher or a friend. These attitudes are reflected in the friendly voice and tone of the agent, as well as the provided positive feedback. In contrast, the non-embodied agent was perceived as emotionless and robotic and correlated the agent to a laptop. This emotional distinction aligns with the study by Lester et al. [35], who states several benefits for agents who can express life-like emotions. According to them, these agents can provide clear problem-solving advice and keep learners highly motivated and engaged. Baylor and Ryu [9], found that the visual appearance of an agent may be important to enhance the experience. Lala and Nishida [33], finally, showed that computer agents can effectively improve communication and boost engagement through realistic verbal and non-verbal communication.

On the other hand, while embodiment is generally expected to enhance perceived realism, some participants described the embodied coxswain as “scary” or “monotone.” Such reactions can be interpreted through the *uncanny valley* [40, 68], which posits that agents approaching human-like appearance without perfect fidelity may evoke discomfort [38]. This suggest that embodiment choices should balance realism with design simplicity to avoid undermining social presence. Thus, the behaviours that we developed for our embodied coxswain may point in a useful direction for making virtual agents in VR sports applications, which we address in the next section.

6.3 Other Uses of Virtual Humans

Our findings show that, even though both agents gave the same positive feedback, the embodied agent was mentioned as giving good feedback throughout the experiments, compared to the non-embodied agent. This observation can well be explained by the research of Graf et al. [24], which posits that positive feedback could cause feelings of relatedness between the player and an embodied coach. Consequently, rowers might have more awareness of the feedback of the embodied agent compared to the non-embodied agent.

Our findings thus provide a starting point for exploring the advantages of deploying embodied coaches who provide positive

feedback and behave naturally. Realism offers other potential benefits, for instance, training within realistic 3D environments has proven to effectively prepare users for real-world scenarios [72]. Besides, observing the agent can help understand and learn certain movements, like catching a ball in basketball. Our findings further resonate with those by Graf et al. [24], which show that adding emotional elements to virtual coaches within exergames (VR games involving physical exercise) can enhance performance and motivation. Specifically, a happy coach contributes to a more positive gaming experience and can increase happiness and perceived competence. Additionally, an embodied agent can enhance feelings of relatedness between the player and the coach. These results are likely caused by the positive feedback and praise provided by the cheerful virtual coach.

6.4 Challenges Stem from the Behaviors of Virtual Agents

Our results showed that the feedback of the agents often seemed insincere, as it did not result directly from their actions. This caused a disconnect between the rowers and the agent which aligns with the study of Covaci et al. [18], who express that personalized feedback could enhance performance, as it allows to take the current performance of the user and help them adjust it for the next trial. Besides, virtual situations can guide users by giving additional information and can easily adapt to different competitive situations.

Our participants also frequently seemed confused during the experiments, specifically when the agents stopped providing a rhythm. This occurrence was remarkable, as the pre-observations made it clear that it is not the role of the coxswain to constantly provide the stroke rhythm since this is the responsibility of the first person in the boat. Due to these observations, in the design phase, it was intentionally chosen not to provide a constant rhythm during the experiment. However, given that many participants had limited experience in rowing, they expected the coxswain to give the rhythm continuously. The more experienced rowers did not express the need for a constant rhythm.

Finally, our findings show that some participants positioned in the front of the boat reported a lack of awareness of the rower behind them. However, the LMER analysis revealed no significant difference in social presence between the front and back positions. This might imply that auditory feedback might have a stronger influence on social presence than visual factors, which requires further research. A factorial design isolating individual embodiment features, such as facial animation, voice tone, and agent position, could reveal which elements most strongly influence perceived social presence [35]. This would guide more targeted design choices for virtual coaches.

7 Conclusions

Our findings provide a proof-of-concept that embodiment can enhance perceived social presence in VR rowing. Given the limited sample, scripted feedback, and short-term exposure, these results should be interpreted as context-specific rather than generalisable to all sports or VR team training scenarios. Through our observations and experimental study, we demonstrated that including an embodied coxswain can significantly enhance the social presence

between two rowers compared to a non-embodied agent within a VR rowing experience. Our findings demonstrate that the influence of virtual embodied agents in rowing is promising, which supports prior work, and therefore future studies hold the potential to extend the use of embodied agents to different exercise fields. Embodied coaches not only enhance social presence but can also be deployed to improve performance, skill development, motivation, and engagement. In that sense, our findings also pave the way for other sports where coaches also play a significant role.

Acknowledgments

This work was funded by the VU Amsterdam–University of Twente Alliance ('Rowing Reimagined' project).

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